Liam Murphy

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Experience

FSAE Aerodynamicist/Test Engineer September 2024 - Present
CU Boulder Racing Team, Boulder, CO

- Designed inlet of the team's first ground-effect undertray, with a CFD calculated 25% downforce gain and 1% drag decrease compared to the car with no undertray
- Manufactured carbon composite aerodynamic elements and integrated them into the final vehicle assembly
- Led suspension potentiometer testing to validate CFD downforce calculations, creating a validation platform for future aero and chassis iterations

Launch System Engineer August 2026 - Present
CU Boulder SuperSpeed, Boulder, CO

- Designing a portable high acceleration bungee launcher for CUSS1, a high speed drone targeting 200+ mph
- Used FEA to design parts with a FOS of 1.5, cutting weight for increased launcher performance
- Routed the bungee through the extrusion rail's central cavity with a multi-pulley system to quadruple effective cord length, sized for an 18 m/s release at an 8° launch angle
- Built a launch calculator to size/tune launch and braking cords based on rail geometry and bungee properties

Surveillance Drone Engineer April 2026 - Present
CU Boulder Robotics, Boulder, CO

- Designing a small quadcopter that spots opposing robots with an onboard camera and relays their positions to the team in real time
- From a 5 m hover the camera sees about 66% of the field, so the drone tracks most robots from one position

Projects

FPV Racing/Cinematic Drones September 2025 - Present

- Built a racing micro quadcopter (3+ target power to weight), iterating the frame from flight/static test data
- Built a 5" deadcat layout cinematic drone, 3D printed carbon filament frame designed using tested flight loads
- Built a mod for the Liftoff FPV simulator to fly my own frame geometry and motors, used to evaluate the unconventional layout and build a preliminary PID tune

Quadcopter Motor Test Stand August 2026 - Present

- Built a low cost thrust stand, motor mounted axially over a 5 kg S-type load cell, read by an ESP32 through an amplifier, with laptop throttle control and data logging
- Tested selected 5" deadcat motor/prop and measured peak thrust at 1.4kgf, which was used to design frame

Education

University of Colorado Boulder May 2027 Graduation
B.S. Aerospace Engineering

- Organizations: CU Boulder Racing Team FSAE Branch, CU Boulder Robotics

Skills

- **Technical:** Aerodynamic Design, CFD, FEA, CAD, MATLAB, Python, Test Instrumentation/DAQ, Drone Design, Composites Manufacturing, 3D Printing
- **Other:** Team Leadership, Project Management, Technical Communication